Battery SOH estimation across chemistries

Collaborative modeling | Few-shot adaptation | Reusable knowledge

01 Context & value

Battery health at scale



E-mobility Energy storage

Battery health assessment



Industry Fleets Research

Distributed data; restricted sharing



LFP NCA NCM

Chemistry shifts challenge transfer

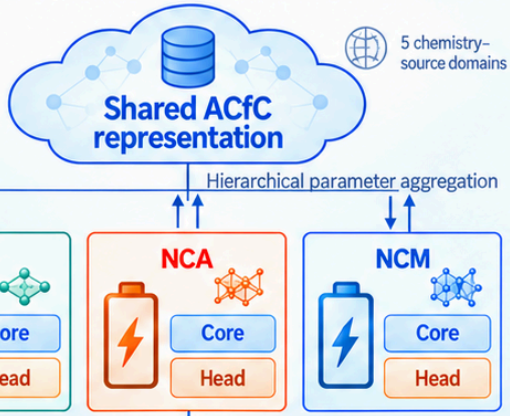
Research value

- Maintenance decisions
- Collaborative learning
- Deployment potential

02 Method & implementation

Hierarchical federated learning

Share common features; retain domain heads



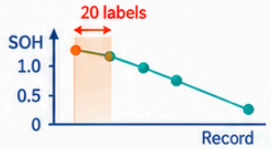
Shared base: simulated federated aggregation

Source mappings are fitted separately

Domain heads and source-specific mappings

Target-cell adaptation

- 20 labelled observations
- Freeze base encoder; calibrate locally



SOH estimation with domain-specific calibration

3 chemistries | 20-shot MAE 0.00935 | 378 cells

03 Reuse & extension

Reusable, adaptable, extensible

Toward reusable collaboration



Extend tasks and chemistries

Reuse features; adapt with few labels

REUSE Features → **ADAPT** Labels → **EXTEND** Domains

External evaluation

NASA / Oxford; bounded reuse

Solid: studied workflow ------→ Dashed: proposed extensions